

Aditya Gola

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EDUCATION

- **ABES Engineering College** Ghaziabad, U.P., India
Bachelor of Technology - Computer Science and Engineering; CGPA: 7.92
2023 - Present
- **The Millennium School** Noida, U.P., India
Senior Secondary; Percentage: 88.8%
2020 - 2021
- **The Millennium School** Noida, U.P., India
Secondary; Percentage: 83.8%
2022 - 2023

TECHNICAL SKILLS

- **Programming Languages:** C++, Java, Python, JavaScript, TypeScript, SQL
- **Frontend Development:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- **Backend Development:** Node.js, Express.js, FastAPI, RESTful API Development
- **AI/ML & Generative AI:** LLM APIs, Retrieval-Augmented Generation (RAG), Hugging Face Transformers, ResNet50, Computer Vision, Image Classification, Image Captioning
- **Data & Visualization:** Pandas, Matplotlib, JSON
- **Databases & Storage:** MySQL, JSON-based Data Management
- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook
- **Core Computer Science:** Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

PROJECTS

- **OmniVision: AI Image Assistant:**
 - **Image Understanding:** Developed an AI-powered image assistant using ResNet50 and Hugging Face Transformers for image classification and semantic labeling.
 - **Automated Metadata:** Designed image captioning and metadata generation to support keyword-based indexing and retrieval.
 - **Image Enhancement:** Built an image enhancement pipeline to improve image quality and resolution before storage and processing.
 - **Content Organization:** Integrated Python-based deep learning workflows to automate image analysis and simplify content organization.
- **CodeLumen AI: Code Review Assistant:**
 - **Code Analysis:** Built a stateless web application using Python and JavaScript that analyzes source code and returns bug detection, plain-English explanations, and improvement suggestions without requiring setup.
 - **Static Analysis:** Implemented an AST-based analysis engine to detect structural issues, including unsafe `eval()` calls and use-before-assignment, beyond simple pattern matching.
 - **Security Scanning:** Integrated NVD CVE-based dependency vulnerability scanning to support automated security analysis.
 - **LLM Integration:** Added an optional LLM-based analysis layer for deeper code explanations while retaining a self-contained core analysis engine.

PUBLICATIONS

- **Carbon footprint detector: an empirical approach for green and clean energy in smart cities** 04/11/2025
International Journal of Environmental Engineering
 - Led the foundational literature review and engineered the core technical framework, contributing to the technical and analytical methodologies used in the published manuscript.

CODING PROFILES

- **HackerRank:** 5-star rating in Python.
- **LeetCode:** Solved 300+ problems focused on data structures and algorithms.

DEVELOPMENT ACTIVITY

- **GitHub:** Develop and maintain programming projects, using Git/GitHub for version control, source-code organization, and project documentation.

CERTIFICATIONS

- **Problem Solving (Intermediate):** HackerRank
- **Python (Basic):** HackerRank

HONORS AND AWARDS

- **ARDEMA Hackathon Winner:** Secured 1st place by developing a Disease Detector tool for early disease prediction using machine learning techniques within a 2-day timeframe.